

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION
Detailed Marking Report — Biology Exam-16

Seemab · Class 9 (Science Group) · Chapters 1 to 7 (Chapter 7 to §7.6 only) · Attempted 13 September 2026 · deep-check verified

34 / 36

TOTAL

94.4%

PERCENTAGE

A+

GRADE

18 / 20

SECTION A

8 / 8

SECTION B

8 / 8

SECTION C

SECTION A — every question (18 / 20)

#	Question	Hers	Key	Mark	Note and source in her textbook
1	Fraction of 100 healthy persons with NO trace of Plasmodium in their blood.	B	B	1	Ch 1, §1.8 Cause of Malaria – Experiments & Results, p.14 — SLO 8 (steps of the biological method; malaria case study data)
2	A statement that describes an observable occurrence in nature which appears always true, but do	D	D	1	Ch 1, §1.7 Hypothesis, Theory and Law, Summary point 13, p.13 — SLO 9 (evaluate hypothesis, theory and law)
3	Bioinformatics is a combination of biology and:	A	A	1	Ch 1, §1.4 Careers in Biology, p.10 — SLO 6 (identify careers in biology, including emerging careers)
4	Correctly written scientific name of the lion, Panthera leo, following binomial nomenclature ru	C	C	1	Ch 2, §2.7 Binomial Nomenclature, p.29 — SLO 7 (rules for writing binomial names)
5	Multicellular heterotrophic eukaryotes lacking a cell wall and chlorophyll, including both inse	A	A	1	Ch 2, §2.4 The Domain Eukarya – Kingdom Animalia, p.25 — SLO 4 (three domains and the kingdoms of Domain Eukarya)
6	Which is NOT one of the three components of the cytoskeleton?	D	D	1	Ch 3, Cytoskeleton, p.37 — SLO 3 (sketch and outline the roles of subcellular organelles)
7	Layer of the plant cell wall formed after the cell stops growing, thickened with lignin:	B	B	1	Ch 3, Cell Wall, p.33 — SLO 3 (sketch and outline the roles of subcellular organelles, including the cell wall)
8	Pinching-in structure formed during cytokinesis in an ANIMAL cell:	C	C	1	Ch 4, §4.3 Cytokinesis, p.52 — SLO 2 (explain mitosis and its stages, by use of sketch and diagrams)
9	A cell with $2n = 14$ undergoes meiosis. Chromosomes per cell at the end of MEIOSIS I:	A	A	1	Ch 4, §4.4 Meiosis, p.54–56 — SLO 2/3 (explain meiosis and its stages; maintenance of chromosome number)
10	A cell with $2n = 10$ replicates its DNA (S phase). Chromosomes/chromatids at late prophase of MI	D	D	1	Ch 4, §4.2 S Phase / §4.3 Prophase, p.48, 50 — SLO 1/2 (cell cycle; stages of mitosis)
11	Skeletal system's role in homeostasis, besides support/protection/movement:	C	C	1	Ch 5, Table 5.2 Major Organ Systems, p.69–70 — SLO 6 (discuss how organs/systems of the body work to maintain homeostasis)
12	Organ system made of leukocytes, tonsils, adenoids, thymus and spleen, that removes disease-cau	B	A	0	Ch 5, Table 5.2 Major Organ Systems, p.69–70 — SLO 6 (discuss how organs/systems of the body work to maintain homeostasis)
13	Muscular system (with the integumentary system) helps maintain homeostasis by:	D	D	1	Ch 5, Table 5.2 Major Organ Systems, p.69–70 — SLO 6 (discuss how organs/systems of the body work to maintain homeostasis)
14	Nitrogen base with a DOUBLE-ring structure:	B	B	1	Ch 6, §6.7 Nucleic Acids – Nucleotide structure, p.87 — SLO 7 (structure of DNA as a double helix made of nucleotides)
15	General property of ALL lipids (insolubility in water), distinct from "saturated/unsaturated" (	A	A	1	Ch 6, §6.4 Lipids, p.82 — SLO 4 (outline the structure, function and sources of lipids)
16	Approximate number of different amino acid types commonly found in proteins:	C	C	1	Ch 6, §6.3.2 Amino Acids, p.79 — SLO 3 (outline the structure of proteins with the structure of amino acids)
17	Covalent bonds between the phosphate groups of ATP, which store releasable energy:	D	B	0	Ch 7, §7.6.2 Structure of ATP Molecule, p.101 — SLO 6 (ATP as the energy currency of the cell)
18	What happens to an enzyme molecule after it converts substrate into product:	D	D	1	Ch 7, §7.2.1 Characteristics of Enzymes – Enzymes are required in small quantity, p.95 — SLO 2 (enzymes as biological catalysts and their characteristics)
19	Series of connected reactions where each enzyme acts on the product of the previous one:	C	C	1	Ch 7, §7.2.1 Characteristics of Enzymes – Enzymes make metabolic pathway, p.95 — SLO 2 (enzymes as biological catalysts and their characteristics)
20	Digestion of food into smaller, simpler molecules is an example of:	B	B	1	Ch 7, §7.1 Metabolism and Its Types, p.94 — SLO 1 (metabolism, catabolism and anabolism)

Red rows lost the mark; amber rows are judgement calls. Every letter was re-read from the scan and cross-checked by an independent deep-check pass.

34

out of 36 · 94.4%

18/20

MCQs · 90%

16/16

Sections B + C

The short version. Seemab scored 34 out of 36. Her written work was full marks again, and the four questions set deliberately from her own past biology mistakes all came back correct. Across Physics exam-15 and this paper, **eight of eight re-set errors are now fixed.**

She lost two multiple-choice marks. One is a genuinely confusable pair in her own textbook (immune system against lymphatic system), the other a single recalled term (high-energy bonds). Neither is a method problem.

All four repeat errors: corrected.

Q	Her earlier mistake	This time
2	Exam-15: shown the definition of a scientific law , answered "theory"	Scientific law ✓
8	Exam-15: gave the plant answer (phragmoplast) to an animal cytokinesis question	Cleavage furrow ✓
14	Exam-01: confused purines and pyrimidines	Guanine (a purine) ✓
15	Exam-15: chose "saturated" where "hydrophobic" was wanted	Hydrophobic ✓

On Q2 she wrote her reasoning, and it shows the confusion is genuinely resolved rather than guessed through: *"hypothesis can be true or false, deduction is using logical terms like if or then, and scientific theory is giving theory after experiment and explains how and why it happens."* She eliminated theory precisely because the stem described something that does **not** explain why.

Section A — 18/20

Q	Topic	Her answer	Correct	Comment
12	Organ system of leukocytes, tonsils, adenoids, thymus, spleen (Table 5.2)	B, Lymphatic	A, Immune	Checked against her book: Table 5.2 lists these under the immune system; the lymphatic row is lymph, lymph nodes and lymph vessels. A fair trap, because the book's lymphatic row also says it "defends against infection and disease".
17	Bonds between ATP's phosphate groups	D, Hydrogen bonds	B, High-energy bonds	Her book, Ch7 p.101: "covalent bonds between phosphate groups are called 'high energy' or 'energy-rich' bonds".

All 20 correct answers were re-derived from the printed stems and her own textbook pages, not from the answer key, and no key error was found. A shift guard gave 18 matches at the correct alignment against 0 to 2 at every other offset.

Section B — 8/8

Q	Topic	Mark	Comment
2(i)	Four levels of protein structure	4/4	Primary (amino-acid sequence, peptide bonds), secondary (α -helices and β -pleated sheets from hydrogen bonding), tertiary (3-D shape held by hydrogen, ionic and disulfide bonds) and quaternary (two or more chains joined into a functional protein). All four correct and more detailed than the book.
2(ii)	Why malaria cannot be eradicated	4/4	Four reasons: Plasmodium's complex life cycle across humans and mosquitoes; rapid mosquito breeding; insecticide resistance in mosquitoes; drug resistance in Plasmodium. Ruled by Atif, 13 Sep: all four credited. See note below.

The 2(ii) ruling. Her first two reasons rest on book content (Plasmodium lives in both mosquito and human; mosquitoes breed in stagnant water). Insecticide and drug resistance are correct biology but appear nowhere in her textbook. The independent second pass recommended 2/4 on that basis, giving 32/36.

The marks were awarded because the defect is in our question rather than her answer. The paper told her to answer "as discussed in the textbook", but the book only poses this as an exercise item and never answers it, and our own answer key's four reasons were inferences too. Her school set this exact question on its First Term paper, so these reasons may reflect what was taught in class. **Under the strict reading the paper is 32/36 (88.9%).**

Both Section B questions were items her school examined that had never appeared in any of our fifteen earlier biology papers.

Section C — 8/8

Part	Required	Her answer	Mark	Comment
(a)	Identify A, B, C on the membrane diagram	A hydrophilic head of a phospholipid; B integral protein; C carbohydrate chain	3/3	Label A's printed line ends on the head of a single phospholipid, so her answer is more precise than the key's "phospholipid bilayer". "Integral protein" is a correct classification of the transport protein, though not the book's own word.
(b)	Chemical composition	Proteins 60–80%, phospholipids 20–40%, traces of carbohydrates	2/2	Word for word from her book, Ch3 p.33.
(c)	What the fluid mosaic model postulates about the arrangement of proteins	A phospholipid bilayer with proteins incorporated in a mosaic fashion; proteins move laterally, making the membrane fluid	3/3	The answer key gave its third mark for cholesterol, which is not about the arrangement of proteins and was not asked. She answered the question as set.

Two defects in our paper

- **Q17 offered "glycosidic bonds" as an option.** That term does not appear anywhere in her Chapter 6 or 7 scans, and it was removed from exam-15 for exactly that reason. Wrong options must also come from her book. It did not affect her mark.
- **Two answer-key problems**, both resolved in her favour above: the Section C(c) cholesterol mark answered a different question, and the 2(ii) key's reasons were not from the book.

What to say to her

- **Every re-set mistake is fixed.** Law against theory, animal cytokinesis, hydrophobic, purines. Together with physics, that is eight out of eight.
- **Written work is full marks again.** The protein-structure answer in particular went beyond what the book asks.
- **Two terms to learn:** the immune system is leukocytes, tonsils, adenoids, thymus and spleen, while the lymphatic system is lymph, lymph nodes and vessels; and the bonds between ATP's phosphates are called high-energy bonds.

Marking independently deep-checked. All MCQ answers re-derived from her textbook, every letter re-read from the scan, shift guard run. The second pass agreed on every question except 2(ii), which was referred to Atif and ruled in her favour.